

Revisiting the effects of maternal education on adolescents' academic performance: Doubly robust estimation in a network-based observational study

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Abstract

In many contexts, particularly when study subjects are adolescents, peer effects can invalidate typical statistical requirements in the data. For instance, it is plausible that a student's academic performance is influenced both by their own mother's educational level as well as that of their peers. Since the underlying social network is measured, the Add Health study provides a unique opportunity to examine the impact of maternal college education on adolescent school performance, both direct and indirect. However, causal inference on populations embedded in social networks poses technical challenges, since the typical no interference assumption no longer holds. While inverse probability-of-treatment weighted (IPW) estimators have been developed for this setting, they are often highly unstable. Motivated by the question of maternal education, we propose doubly robust (DR) estimators combining models for treatment and outcome that are consistent and asymptotically normal if either model is correctly specified. We present empirical results that illustrate the DR property and the efficiency gain of DR over IPW estimators even when the treatment model is misspecified. Contrary to previous studies, our robust analysis does not provide evidence of an indirect effect of maternal education on academic performance within adolescents' social circles in Add Health.

1 Introduction

Increasingly, there is a marked interest in estimating causal effects under interference, which arises when the potential outcome of an individual is affected by the treatment, exposure, or intervention of another. In classical causal inference, interference is often assumed away in favour of the stable unit treatment value assumption (SUTVA) (Rubin, 1974). However, we can think of numerous settings where this assumption is likely violated and where interventions not only have direct effects on recipients but also *spill over* onto their close contacts, hence generating indirect (spillover) effects. For instance, if we consider a vaccine trial on a population embedded in a network, chances are that the vaccination status of one individual will spill over onto a neighbour and reduce their risk of infection (Liu et al., 2019; Ogburn & VanderWeele, 2017; Perez-Heydrich et al., 2014). Outside the realm of infectious disease, another area where interference naturally manifests is education, where peer effects or *spillover* effects have been extensively studied, be it in the context of an experiment (Basse & Feller, 2018; Hong & Raudenbush, 2006) or a nonrandomized exposure such as maternal education (Bifulco et al., 2011; Fletcher et al., 2020). Indeed, it is this latter question that motivates this work: we wish to disentangle the direct and indirect effects of maternal education on adolescent school performance.

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The bulk of the interference literature has focused on relaxing the SUTVA assumption to allow for arbitrary interference within disjoint clusters, a setting known as *partial interference* (Hudgens & Halloran, 2008; Liu & Hudgens, 2014; Perez-Heydrich et al., 2014; Sobel, 2006; Tchetgen & VanderWeele, 2012). In this framework, independent groups of units are used to define the interference sets and information on shared connections within each cluster is typically not measured or utilized (Lee et al., 2023). Liu et al. (2016) developed an inverse probability-of-treatment weighted (IPW) estimator for causal effects under a general interference framework, which allowed for overlapping interference sets. This paved the way for methodology for observational network data under the counterfactual framework.

Recently, singly robust inference methods have been proposed for network-based observational studies where the interference graph is limited to first-order neighbours (Forastiere et al., 2021; Lee et al., 2023). Forastiere et al. (2021) rely on a propensity score regression approach, which requires modelling assumptions regarding the dependence of the outcome on the covariates. Drawing upon the work of Liu et al. (2016) in the general interference setting, Lee et al. (2023) proposed IPW estimators for the direct and indirect effects of a treatment under network interference and derived a variance estimator that allows for overlapping interference sets. However, IPW estimators are known to be highly unstable and to have large variances, especially in the presence of extreme propensity scores (close to either 0 or 1) which are more likely to arise in the interference setting (Liu et al., 2019).

Doubly robust (DR) estimators present certain advantages compared to IPW estimators. They combine an outcome model and a treatment model and produce a consistent estimator of the causal effect if either model has been correctly specified, hence providing some protection against model misspecification (Liu et al., 2019). Furthermore, DR estimators are known to have higher efficiency than IPW estimators, even when only the treatment nuisance model is correctly specified. Several estimators with the double-robust property have been proposed in the no interference setting (Bang & Robins, 2005, 2008; Scharfstein et al., 1999); see Kang and Schafer (2007) for an extensive review. Liu et al. (2019) proposed extensions of these estimators in the partial interference framework. Extensions of the DR estimators of Liu et al. (2019) have yet to be developed for observational network data under the counterfactual framework. While Ogburn et al. (2022) proposed a targeted maximum likelihood estimator (TMLE) for observational social network data, their approach targets a single connected network component and is not specifically designed for multilevel network data, which is prevalent in educational research.

An issue inherent to network-based observational studies is the presence of homophily, which is defined as the increased tendency of units with similar characteristics of forming ties. In our study of maternal education, we expect latent variable dependence due to latent traits that both drive the formation of friendships and predict an individual's socio-economic background. If these characteristics are observed, then observations can be rendered conditionally independent (Ogburn et al., 2022). While it would be unrealistic to think that we could measure all traits driving homophily, latent variable dependence poses a threat to the identification of causal effects (Forastiere et al., 2018). Frameworks proposed by Forastiere et al. (2021) and Lee et al. (2023), which we draw upon, require neighbours' treatments to be conditionally independent given observed covariates. Given the possibility of latent treatment dependence, it appears even more important to develop inference methods that do not solely rely upon a correctly specified treatment model.

Our article addresses an important methodological gap by extending the DR estimators proposed by Liu et al. (2019) in the partial interference setting to the more general context of network-based observational studies. We propose two novel estimators where the interference set is defined as the set of first-order neighbours in a sociometric network. By design, these estimators take into account the interconnections among individuals' treatments and outcomes. Under certain conditions, they can be shown to be consistent and asymptotically normal if either model, but not necessarily both, is correctly specified, assuming that the network can be written as the union of disjoint subgraphs. The first estimator, a regression estimator with residual bias correction, is endowed with the double robustness property whether or not the outcome has a multilevel structure. The second estimator, a regression estimator with inverse-propensity weighted coefficients, can be shown to be doubly robust if the outcome does not follow a hierarchical model. Using this framework, we apply M-estimation theory to propose appropriate asymptotic variance estimators. We also present empirical evidence of the double robustness property, as well as the efficiency gain of proposed estimators over IPW estimators, even when the treatment model is correctly specified.

We make another contribution by illustrating the threat of latent treatment homophily to the identification of causal effects and show how DR estimation can guard against it.

Studies suggest that an adolescent's academic outcomes (e.g. grade point average, retention) are positively affected not only by their own home environment but also by that of their friends (Bifulco et al., 2011; Fletcher et al., 2020). For instance, the higher the proportion of friends or classmates with college-educated mothers, the greater the likelihood of finishing school and the higher the grades were according to their analyses. In our study leveraging data from Add Health, the direction of the effects found using our proposed causal inference methods supports conclusions drawn by previous authors, although the DR analysis did not achieve statistical significance. Thus, our study does not provide evidence that the composition of friendship circles has an influence on learning outcomes. However, an important caveat is that we had to use a smaller subset of the dataset for computational reasons, potentially leading to an underpowered analysis. The validity of our causal analysis depends on, among other things, either a correctly specified model for the exposure (maternal college education) or a correctly specified model for the grade point average.

The article is structured as follows. In Section 2, we introduce our motivating example using Add Health data which examines the direct and indirect effect of maternal education on school performance. In Section 3, we introduce the potential counterfactual framework under network interference and define relevant causal estimands. In Section 4, we describe different inference strategies: inverse-probability-weighting estimation, regression estimation, and DR estimation. In Section 5, we assess finite sample properties of proposed estimators through simulation studies and illustrate the double robustness property of DR estimators. Having derived new, efficient, and valid tools for estimation, we demonstrate the application of our proposed approach to estimate spillover effects of maternal college education on adolescent school performance using the Add Health data in Section 6. We close with a general discussion in Section 7.

2 Motivating context: peer effects in education

The National Longitudinal Study of Adolescent Health (Add Health) is a longitudinal study of a nationally representative sample of 90,118 adolescent students in grades 7–12 in the U.S. in 1994–95 who were followed throughout adolescence. In addition to socio-demographic and academic achievement characteristics, the survey collected data on friendship nominations through in-school questionnaires administered directly to students, from which a social network can be constructed. We considered an undirected network based on the data on friendship nominations from the first wave of data collection in 1994–95 of Add Health. We defined a friendship as a symmetric relationship, meaning that there was an edge between students i and j if student i listed j as a friend in the in-school survey, or j listed i as a friend, or both. The left side of Figure 1 represents one school in the Add Health network, whereas the right side displays relations among 139 schools, where an edge between two schools is indicative of at least one across-school friendship. A critical assumption is that the network is fully known and observed. Our methodology also assumes that the network can be decomposed into smaller disjoint subgraphs. This assumption is amenable to multilevel network data, which are prevalent in educational research, in which case the disjoint subgraphs correspond to different schools, classrooms, or simply network components (Abacioglu et al., 2019; van Rijsewijk et al., 2018; Stewart et al., 2019).

As peer influence is an ongoing theme in educational research, the Add Health data provide a unique opportunity to examine the causal effect of peers on academic success. Leveraging the Add Health data, Bifulco et al. (2011) found that having a higher proportion of classmates with college-educated mothers was associated with a lower risk of dropping out and a higher likelihood of attending college. Recently, Fletcher et al. (2020) made use of the friendship nominations data and found that having a greater proportion of friends with college-educated mothers increases Grade Point Average (GPA) among female students, but not among male students. Recent methodological developments in causal inference with interference allow us to re-analyse this question under a counterfactual framework. Specifically, we aim to quantify the (direct) effect of having a college-educated mother on GPA, as well as the (indirect) effect of having a higher proportion of friends with college-educated mothers on GPA. Our proposed estimator combines a model for the propensity of maternal education among adolescents' social circles and a model for the outcome, GPA. Throughout the text, we will refer to maternal education as a (self-selected) treatment.

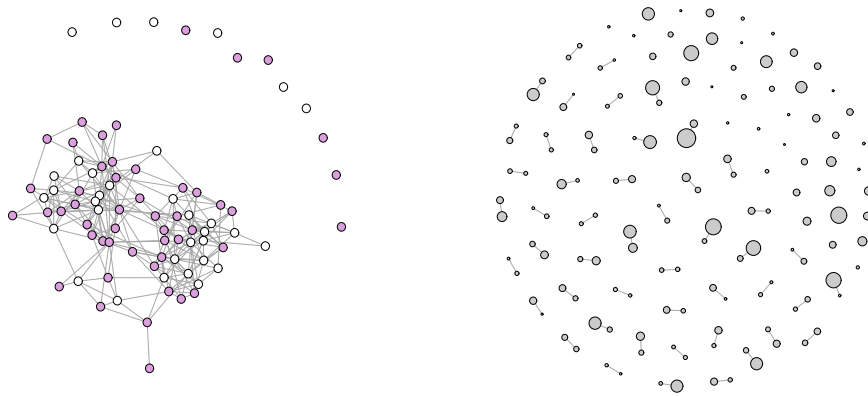


Figure 1. Left: One school of 83 students in the Add Health network, where an edge between two students indicates that either one nominated the other as a friend and plum-coloured nodes represent students with college-educated mothers. Right: Network formed by 139 schools in Add Health, where an edge is present if two schools share at least one friendship; and the size of each node is proportional to the size of the school.

3 Notation, assumptions, and causal estimands

Consider a binary undirected network $G = (\mathcal{N}, \mathbb{E})$, where $\mathcal{N} = \{1, 2, \dots, N\}$ is a set of N nodes and $\mathbb{E}$ is a set of edges with generic element $\{i, j\} = \{j, i\}$ denoting the presence of an edge between nodes i and j . For convenience, we represent the network G by the sociomatrix A , which has entries $A_{ij} = 1$ if $\{i, j\} \in \mathbb{E}$ and 0 otherwise. Since we assume that there are no self-loops, the diagonal elements of A are identically 0. We define the neighbourhood $\mathcal{N}_i$ of node i , $i = 1, 2, \dots, N$, as the set of nodes j which satisfy $A_{ij} = 1$ and the cardinality of $\mathcal{N}_i$ is denoted as $|\mathcal{N}_i| = d_i$. Additionally, we denote $\mathcal{N}_i^* = \mathcal{N}_i \cup i$ as the union of the node i and the nodes in $\mathcal{N}_i$. Thus, as put by Forastiere et al. (2021), it follows that to each node i is associated a partition of $\mathcal{N}$, that is $\{i, \mathcal{N}_i, \mathcal{N}_{-i}\}$, where $\mathcal{N}_{-i} = \mathcal{N} \setminus \mathcal{N}_i^*$.

A graph $H = (\mathcal{N}_H, \mathbb{E}_H)$ is a subgraph of G if $\mathcal{N}_H \subseteq \mathcal{N}$ and $\mathbb{E}_H \subseteq \mathbb{E}$. Further, a subgraph of a graph G is said to be *connected* if there exists a path for any two nodes in the subgraph (Kolaczyk, 2009). We will assume that the network can be expressed as the union of m components $C_1, C_2, \dots, C_m$, where a component of a graph is defined as a maximally connected subgraph (i.e. a connected subgraph that is not part of any larger subgraph) (Kolaczyk, 2009). Attributes measured on the nodes in disjoint subgraph are viewed as a random sample from a super-population and constitute independent data units. This assumption is necessary to introduce regression or model-based estimators of causal effects and to derive asymptotic properties of the proposed estimators.

For each node i , $i = 1, 2, \dots, N$, we observe the node-level vector of attributes $O_i = (X_i, Z_i, Y_i)$, where $X_i = (X_{i1}, \dots, X_{ip})'$ denotes a p -dimensional vector of pretreatment covariates, Z_i denotes a binary (self-selected) treatment with $Z_i = 1$ if treated and 0 otherwise, and Y_i is a univariate outcome of interest. Let $\mathbf{X}_{\mathcal{N}_i} = (X_{j_1}, \dots, X_{j_{d_i}})'$ and $\mathbf{Z}_{\mathcal{N}_i} = (Z_{j_1}, \dots, Z_{j_{d_i}})'$ denote the neighbourhood matrix of pretreatment covariates and neighbourhood vector of treatments for node i , respectively, where $j_k \in \mathcal{N}_i$, $k = 1, \dots, d_i$. Throughout this article, we will use the terms node and individual interchangeably.

In accordance with the Neyman–Rubin counterfactual framework (Rubin, 1974), let $y_i(z_i, \mathbf{z}_{\mathcal{N}_i}, \mathbf{z}_{\mathcal{N}_{-i}})$ denote the potential outcome for individual i if, possibly contrary to fact, they receive treatment value z_i , their neighbourhood receives $\mathbf{z}_{\mathcal{N}_i}$ and the nodes outside of the neighbourhood receive $\mathbf{z}_{\mathcal{N}_{-i}}$. Later, we will assume spillover only from the nodes in $\mathcal{N}_i$ and write the i th potential outcome as $y_i(z_i, \mathbf{z}_{\mathcal{N}_i})$. Note that $z \in \{0, 1\}$ and $\mathbf{z}_{\mathcal{N}_i} \in \mathcal{Z}(d_i)$, where $\mathcal{Z}(d_i)$ denotes the set of vectors of possible treatment allocations of length d_i with $|\mathcal{Z}(d_i)| = 2^{d_i}$. To ensure identifiability of causal effects, we first make the consistency assumption, which states that there cannot be multiple versions of a treatment.

Assumption 1 (Consistency). For $i \in \mathcal{N}$, $Y_i = y_i(Z_i, \mathbf{Z}_{\mathcal{N}_i}, \mathbf{Z}_{\mathcal{N}_{-i}})$.

We further assume the potential outcome of individual i depends on their first-order neighbours' treatments only through a function of the neighbourhood treatment, which is referred

to as the neighbourhood (Forastiere et al., 2021) or stratified interference assumption (Lee et al., 2023).

Assumption 2 (Stratified interference). For $i \in \mathcal{N}$, $\forall z_{\mathcal{N}_{-i}}, z'_{\mathcal{N}_{-i}}$, and $\forall z_{\mathcal{N}_i}, z'_{\mathcal{N}_i}$ such that $\phi(z_{\mathcal{N}_i}) = \phi(z'_{\mathcal{N}_i})$ given a function $\phi: \{0, 1\}^{d_i} \rightarrow \Phi_i$, then,

$$y_i(z_i, z_{\mathcal{N}_i}, z_{\mathcal{N}_{-i}}) = y_i(z_i, z_{\mathcal{N}_i}, z'_{\mathcal{N}_{-i}}).$$

In other words, we assume that the potential outcome of an individual depends on their neighbourhood through a summary of neighbourhood treatment vector, $\phi(z_{\mathcal{N}_i})$. In this article, we will focus on the transformation $\phi: \{0, 1\}^{d_i} \rightarrow \Phi_i$ defined by $\phi(z_{\mathcal{N}_i}) = \sum_{j \in \mathcal{N}_i} z_j$, where $\Phi_i \in \{0, 1, \dots, d_i\}$, which corresponds to the number of treated neighbours. We will denote this quantity $\Sigma z_{\mathcal{N}_i}$ from this point onward. In the context of this case study, this implies that we assume that the GPA of a student is only affected by the home environment (as captured by maternal education) of their peers through the number of friends with a college-educated mother. This device is introduced to reduce the number of potential outcomes from 2^{d_i+1} (if ϕ was the identity function) to $2d_i$. Because of Assumption 2, we can drop the dependence of $y_i(\cdot)$ on the nodes in $\mathcal{N}_{-i}$ and write more simply $y_i(z_i, z_{\mathcal{N}_i})$.

We let $f(z_i, z_{\mathcal{N}_i} | X_i, X_{\mathcal{N}_i}) = P(Z_i = z_i, Z_{\mathcal{N}_i} = z_{\mathcal{N}_i} | X_i, X_{\mathcal{N}_i})$ denote the probability of observing the joint treatment $(z_i, z_{\mathcal{N}_i})$ conditional on individual and neighbourhood covariates, which is also referred to as the propensity score. The following assumptions are extensions of the typical no unmeasured confounding and positivity assumptions. The no unmeasured confounding assumption states that for all $i \in \mathcal{N}$, the covariates X_i are a sufficient conditioning set, such that groups defined by levels of the joint exposure $(Z_i, Z_{\mathcal{N}_i})$ are exchangeable conditionally on X_i . The positivity assumption ensures that there is appropriate covariate overlap across different treatment groups.

Assumption 3 (Conditional exchangeability). For all $i \in \mathcal{N}$, $z_i \in \{0, 1\}$, $z_{\mathcal{N}_i} \in \{0, 1\}^{d_i}$, $y_i(z_i, z_{\mathcal{N}_i}) \perp Z_i, Z_{\mathcal{N}_i} | X_i$.

Assumption 4 (Positivity). For all $i \in \mathcal{N}$ and for all $z_i, z_{\mathcal{N}_i}, X_i, X_{\mathcal{N}_i}$,

$$f(z_i, z_{\mathcal{N}_i} | X_i, X_{\mathcal{N}_i}) > 0.$$

To define the causal estimands of interest, we adopt the Bernoulli treatment allocation standardization proposed by Tchetgen and VanderWeele (2012) in the context of partial interference and Liu et al. (2016) in the general interference setting. That is, average potential outcomes are standardized according to a reference population in which individuals independently receive the exposure (e.g. maternal education) with probability α . Our framework only differs in the fact that we marginalize the potential outcomes over the number of treated neighbours $\Sigma z_{\mathcal{N}_i}$ instead of the neighbourhood treatment vector $z_{\mathcal{N}_i}$ to remain consistent with Assumption 2. Let $\pi(\Sigma z_{\mathcal{N}_i}; \alpha) = \binom{d_i}{\Sigma z_{\mathcal{N}_i}} \alpha^{\Sigma z_{\mathcal{N}_i}} (1 - \alpha)^{d_i - \Sigma z_{\mathcal{N}_i}}$ denote the probability of node i receiving neighbourhood treatment $\Sigma z_{\mathcal{N}_i}$ under Bernoulli allocation strategy α . Similarly, under counterfactual scenario α , the probability of individual i and their neighbourhood jointly receiving exposure $(z_i, \Sigma z_{\mathcal{N}_i})$ is given by $\pi(z_i, \Sigma z_{\mathcal{N}_i}; \alpha) = \alpha^{z_i} (1 - \alpha)^{1 - z_i} \binom{d_i}{\Sigma z_{\mathcal{N}_i}} \alpha^{\Sigma z_{\mathcal{N}_i}} (1 - \alpha)^{d_i - \Sigma z_{\mathcal{N}_i}}$. Define the i th individual's potential outcome conditional on individual exposure z and allocation strategy α as

$$\bar{y}_i(z; \alpha) = \sum_{\Sigma z_{\mathcal{N}_i}=0}^{d_i} y_i(z, \Sigma z_{\mathcal{N}_i}) \pi(\Sigma z_{\mathcal{N}_i}; \alpha),$$

and the marginal individual potential outcome as

$$\bar{y}_i(\alpha) = \sum_{\substack{z_i \in \{0,1\} \\ \Sigma z_{\mathcal{N}_i} \in \{0,1,\dots,d_i\}}} y_i(z_i, \Sigma z_{\mathcal{N}_i}) \pi(z_i, \Sigma z_{\mathcal{N}_i}; \alpha),$$

for $i = 1, 2, \dots, N$. Similar to [Liu et al. \(2019\)](#), we define the population average potential outcome as $\bar{y}(z; \alpha) = m^{-1} \sum_{v=1}^m N_v^{-1} \sum_{i \in C_v} \bar{y}_i(z; \alpha)$ and the marginal average potential outcome as $\bar{y}(\alpha) = m^{-1} \sum_{v=1}^m N_v^{-1} \sum_{i \in C_v} \bar{y}_i(\alpha)$, where $N_v = |C_v|$ is the number of nodes in component C_v . These definitions of the population average potential outcomes differ from those used by [Liu et al. \(2016\)](#) and [Lee et al. \(2023\)](#), who considered a global average of individual potential outcomes rather than an average of group-level averages. Finally, let $\mu_{z,\alpha} = \mathbb{E}[\bar{y}(z; \alpha)]$ and $\mu_\alpha = \mathbb{E}[\bar{y}(\alpha)]$ denote the average potential outcome and the marginal average potential outcome, respectively, where $\mathbb{E}[\cdot]$ denotes the expected value in the super-population. The average potential outcome is an expectation over the marginal distribution of the covariates $\mathbf{X}$ whereas the marginal average potential outcome takes an additional expectation over the individual treatment allocation $\mathbf{Z}$.

Following [Hudgens and Halloran \(2008\)](#) and [Liu et al. \(2016\)](#), we consider different contrasts of the average potential outcomes to define the causal estimands. The direct exposure effect is defined as the difference between the average potential outcomes of untreated and treated individuals for a fixed allocation strategy α , that is, $DE(\alpha) = \mu_{1,\alpha} - \mu_{0,\alpha}$. The spillover effect, also referred to as the *indirect effect*, is defined as the difference between the average potential outcomes under counterfactual scenarios α and α' among the untreated, i.e. $IE(\alpha, \alpha') = \mu_{0,\alpha} - \mu_{0,\alpha'}$. For instance, in the context of our motivating question, the spillover effect corresponds to the expected increment in GPA when changing the counterfactual probability of having a college-educated mother from α' to α among students whose mother has not completed a 4-year college degree. It is possible to define a spillover effect for the treated individuals as well. For allocation strategies α and α' , we also define the total effect $TE(\alpha, \alpha') = \mu_{1,\alpha} - \mu_{0,\alpha'}$ as well as the overall effect $OE(\alpha, \alpha') = \mu_\alpha - \mu_{\alpha'}$. As the name suggests, $TE(\alpha, \alpha')$ can be interpreted as the total effect of treatment, which encompasses both the direct and the spillover effect, since $TE(\alpha, \alpha') = DE(\alpha) + IE(\alpha, \alpha')$.

4 Inferential procedures

4.1 IPW and regression estimators

IPW is commonly used to address confounding in observational studies. By weighting each observation by the inverse probability-of-treatment, we create a pseudo-population in which the covariates are no longer predictive of the treatment, and thus from which a causal effect can be consistently estimated. In the present case study, the procedure consists of weighting each observation by the joint propensity score so as to balance the covariates across groups defined by the maternal education assignment vector in a given neighbourhood (Z_i, Z_{N_i}) . We consider the following IPW estimators of $\mu_{z,\alpha}$ and μ_α ([Lee et al., 2023](#); [Liu et al., 2019](#)):

$$\hat{Y}^{\text{IPW}}(z, \alpha) = \frac{1}{m} \sum_{v=1}^m \frac{1}{N_v} \sum_{i \in C_v} \frac{y_i(Z_i, \Sigma Z_{N_i}) \mathbb{1}(Z_i = z) \pi(\Sigma Z_{N_i}; \alpha)}{\binom{d_i}{\Sigma Z_{N_i}} f(Z_i, Z_{N_i} | \mathbf{X}_i, \mathbf{X}_{N_i})}, \quad (1)$$

$$\hat{Y}^{\text{IPW}}(\alpha) = \frac{1}{m} \sum_{v=1}^m \frac{1}{N_v} \sum_{i \in C_v} \frac{y_i(Z_i, \Sigma Z_{N_i}) \pi(Z_i, \Sigma Z_{N_i}; \alpha)}{\binom{d_i}{\Sigma Z_{N_i}} f(Z_i, Z_{N_i} | \mathbf{X}_i, \mathbf{X}_{N_i})}. \quad (2)$$

The estimators in (1) and (2) require a model for the propensity score. We introduce the following assumption, which states that all variables driving homophily in the treatment are included in the vector of pre-treatment covariates $\mathbf{X}_i$, $\forall i \in \mathcal{N}$. This assumption was made by previous authors ([Forastiere et al., 2021](#); [Lee et al., 2023](#)), although not explicitly.

Assumption 5 (No latent treatment homophily). For all $i \in \mathcal{N}$, conditional on a component-level random effect b_v , $v = 1, \dots, m$,

$$P(Z_i = z_i, Z_{N_i} = z_{N_i} | \mathbf{X}_i, \mathbf{X}_{N_i}, b_v) = \prod_{j \in \{i, N_i\}} P(Z_j = z_j | \mathbf{X}_j, b_v).$$

Thus, if we adopt a mixed effects logistic regression model for the treatment, the propensity score is assumed to take the form

$$f(Z_i, Z_{N_i} | X_i, X_{N_i}; \gamma) = \int_{-\infty}^{\infty} \prod_{j \in \{i, N_i\}} p_j^{Z_j} (1 - p_j)^{1-Z_j} f(b_v) db_v, \quad (3)$$

where $b_v \sim N(0, \phi_b)$ is a component-level random effect, $\gamma = (\gamma_0, \gamma_1, \dots, \gamma_p)^\top$ is the $(p+1)$ -dimensional parameter vector of the propensity score model, and $p_j = P(Z_j = 1 | X_j, b_v) = \text{logit}^{-1}(\gamma_0 + \gamma_{1:p}^\top X_j + b_v)$, where $\gamma_{1:p} = (\gamma_1, \dots, \gamma_p)^\top$. The component-level random effect accounts for potential dependence among the maternal education indicators of students within each component. The IPW estimators can thus be computed using $f(Z_i, Z_{N_i} | X_i, X_{N_i}; \hat{\gamma}, \hat{\phi}_b)$, where $(\hat{\gamma}, \hat{\phi}_b)$ is the maximum likelihood estimator of (γ, ϕ_b) . Estimators in (1) and (2) differ from those of Lee et al. (2023) in that they assume a priori that the potential outcomes of individual i only depends on the exposures of other individuals through the number of exposed nearest neighbours Σz_{N_i} , so that $Y_i = y_i(Z_i, Z_{N_i}) = y_i(Z_i, \Sigma z_{N_i})$. Estimators in (1) and (2) can be shown to be unbiased for the true average potential outcomes conditional on z and allocation strategy α when the propensity scores are known (see [online supplementary material, Appendix F.1](#)). When (γ, ϕ_b) is estimated from the observed data, Lee et al. (2023) showed that the IPW estimators are consistent and asymptotically normal under Assumptions 1–5 and assuming a correctly specified mixed effects logistic regression model for $f(Z_i, Z_{N_i} | X_i, X_{N_i}; \gamma, \phi_b)$.

We could alternatively estimate the average potential outcomes via outcome modelling, thereby extending the regression estimators introduced by Liu et al. (2019) to the network setting. Direct standardization may be used to identify causal effects, since Assumptions 1 and 3 imply

$$\mathbb{E}[y_i(z_i, z_{N_i}) | X_i] = \mathbb{E}[y_i(z_i, z_{N_i}) | Z_i = z_i, Z_{N_i} = z_{N_i}, X_i] = \mathbb{E}[Y_i | Z_i = z_i, Z_{N_i} = z_{N_i}, X_i].$$

Thus, this suggests fitting a parametric outcome regression model for GPA such as $m_i(Z_i, \Sigma z_{N_i}, X_i; \beta) = \mathbb{E}[Y_i | Z_i, Z_{N_i}, X_i] = \beta_0 + \beta_Z Z_i + \beta_{Z_N} h(\Sigma z_{N_i}) + \sum_{k=1}^p \beta_k X_{ik}$. The parameter estimate $\hat{\beta}$ may then be used to compute the standardized mean outcome from the predicted outcome values, leading to the following regression estimators of $\mu_{z,\alpha}$ and μ_α :

$$\hat{Y}^{\text{REG}}(z, \alpha) = \frac{1}{m} \sum_{v=1}^m \frac{1}{N_v} \sum_{i \in C_v, \Sigma z_{N_i}=0}^{d_i} m_i(z, \Sigma z_{N_i}, X_i; \hat{\beta}) \pi(\Sigma z_{N_i}; \alpha), \quad (4)$$

$$\hat{Y}^{\text{REG}}(\alpha) = \frac{1}{m} \sum_{v=1}^m \frac{1}{N_v} \sum_{i \in C_v} \sum_{\substack{z_i \in \{0,1\} \\ \Sigma z_{N_i} \in \{0,1,\dots,d_i\}}} m_i(z_i, \Sigma z_{N_i}, X_i; \hat{\beta}) \pi(z_i, \Sigma z_{N_i}; \alpha). \quad (5)$$

It can be shown that the estimators in (4) and (5) are consistent and asymptotically normal using standard estimating equation theory if the outcome regression model is correctly specified.

Estimators of the direct, indirect, total, and overall effect are then defined as $\widehat{DE}(\alpha) = \hat{Y}(1, \alpha) - \hat{Y}(0, \alpha)$, $\widehat{IE}(\alpha, \alpha') = \hat{Y}(0, \alpha) - \hat{Y}(0, \alpha')$, $\widehat{TE}(\alpha, \alpha') = \hat{Y}(1, \alpha) - \hat{Y}(0, \alpha')$ and $\widehat{OE}(\alpha, \alpha') = \hat{Y}(\alpha) - \hat{Y}(\alpha')$, respectively, where $\hat{Y}(\cdot)$ is either the IPW or the regression estimator of the average potential outcome. The IPW estimators of the causal effects are consistent if the model for the treatment is correctly specified. Likewise, the regression estimators are consistent if the outcome model is correctly specified. In the next section, we introduce DR estimators, which have the appealing property of being consistent if either the treatment or the outcome model is correctly specified.

4.2 Regression estimation with residual bias correction

Following the definitions of inverse-probability weighting and regression estimators, we propose the bias corrected DR (DR-BC) estimators for the average potential outcomes $\mu_{z,\alpha}$ and μ_α ,

which are given by $\hat{Y}^{\text{DR-BC}}(z, \alpha) = m^{-1} \sum_{v=1}^m \hat{Y}_v^{\text{DR-BC}}(z, \alpha)$ and $\hat{Y}^{\text{DR-BC}}(\alpha) = m^{-1} \sum_{v=1}^m \hat{Y}_v^{\text{DR-BC}}(\alpha)$, respectively, where

$$\hat{Y}_v^{\text{DR-BC}}(z, \alpha) = \frac{1}{N_v} \sum_{i \in C_v} \left\{ \sum_{\Sigma z_{N_i}=0}^{d_i} m_i(z, \Sigma z_{N_i}, X_i; \hat{\beta}) \pi(\Sigma z_{N_i}; \alpha) + \frac{\mathbb{1}(Z_i = z)}{\binom{d_i}{\Sigma z_{N_i}} f(Z_i, Z_{N_i} | X_i, X_{N_i}; \hat{\gamma}, \hat{\phi}_b)} \{y_i(Z_i, \Sigma z_{N_i}) - m_i(Z_i, \Sigma z_{N_i}, X_i; \hat{\beta})\} \pi(\Sigma z_{N_i}; \alpha) \right\}, \quad (6)$$

and

$$\hat{Y}_v^{\text{DR-BC}}(\alpha) = \frac{1}{N_v} \sum_{i \in C_v} \left\{ \sum_{\substack{z_i \in [0,1] \\ \Sigma z_{N_i} \in \{0,1,\dots,d_i\}}} m_i(z_i, \Sigma z_{N_i}, X_i; \hat{\beta}) \pi(z_i, \Sigma z_{N_i}; \alpha) + \frac{y_i(Z_i, \Sigma z_{N_i}) - m_i(Z_i, \Sigma z_{N_i}, X_i; \hat{\beta})}{\binom{d_i}{\Sigma z_{N_i}} f(Z_i, Z_{N_i} | X_i, X_{N_i}; \hat{\gamma}, \hat{\phi}_b)} \pi(Z_i, \Sigma z_{N_i}; \alpha) \right\}. \quad (7)$$

Estimators in (6) and (7) are extensions of the DR estimators proposed by Liu et al. (2019) in the partial interference setting, which are themselves motivated by the estimators proposed by Scharfstein et al. (1999) for the setting where there is no interference. The DR estimators with residual bias correction combine the regression estimator and the inverse weighted residuals of the regression estimator. They differ from those of Liu et al. (2019) in that they make use of network information by allowing each individual to have their own interference set. The DR property can be grasped informally as follows. When the outcome model is correctly specified, the regression estimator converges to $\mu_{z,\alpha}$ since the expected value of the inverse weighted residuals is 0. On the other hand, when the treatment model is correctly specified while the outcome model is not, the second term consistently estimates the bias of the first term (Kang & Schafer, 2007).

To characterize the large-sample distribution of the DR estimators, we leverage the asymptotic framework proposed by Liu et al. (2016) in the general interference setting, of which network interference is a special case. Recall that we assumed that the observed network G could be divided up into m components or disjoint subgraphs $C_1, C_2, \dots, C_m$, where the observable random variables O_v from different components are independent and identically distributed. We write the observed outcome of the i th node in component v as Y_{vi} and the vector of observed outcomes as $\tilde{Y}_v = (Y_{v1}, \dots, Y_{vN_v})^\top$. Similarly, let X_{vi} and Z_{vi} denote the observed covariates and treatment for individual i in component v and define the corresponding component-level vectors $\tilde{X}_v$ and $\tilde{Z}_v$. Under the above assumptions, we have that $O_v = (\tilde{X}_v, \tilde{Z}_v, \tilde{Y}_v)$ are independently and identically distributed for $v = 1, \dots, m$. Let F denote the cumulative distribution function of O_v . Again, we make the consistency assumption, i.e. $Y_{vi} = y_{vi}(Z_{vi}, Z_{vN_i})$, where Z_{vN_i} denotes the neighbourhood vector of treatments for node i in component v . Finally, we also make the exchangeability assumption, that is, $y_{vi}(z_i, z_{N_i}) \perp Z_{vi}, Z_{vN_i} | X_i$. Let $\psi_{za}^{\text{DR-BC}}(O_v; \mu_{za}, \beta, \gamma, \phi_b) = \hat{Y}_v^{\text{DR-BC}}(z, \alpha) - \mu_{za}$ and $\psi_\beta(O_v; \beta)$, $\psi_\gamma(O_v; \gamma, \phi_b)$, and $\psi_{\phi_b}(O_v; \gamma, \phi_b)$ denote the estimating functions corresponding to $\hat{\beta}$, $\hat{\gamma}$, and $\hat{\phi}_b$ such that $\hat{\theta}^{\text{DR-BC}} = (\hat{Y}^{\text{DR-BC}}(0, \alpha), \hat{Y}^{\text{DR-BC}}(1, \alpha), \hat{\beta}, \hat{\gamma}, \hat{\phi}_b)^\top$ solves the estimating equation

$$\psi^{\text{DR-BC}}(\theta) = \sum_{v=1}^m \psi^{\text{DR-BC}}(O_v; \theta) = 0,$$

where $\theta = (\mu_{0a}, \mu_{1a}, \beta, \gamma, \phi_b)^\top$ and

$$\begin{aligned} \psi^{\text{DR-BC}}(\mathbf{O}; \theta) = & (\psi_{0a}^{\text{DR-BC}}(\mathbf{O}; \mu_{0a}, \beta, \gamma, \phi_b), \psi_{1a}^{\text{DR-BC}}(\mathbf{O}; \mu_{1a}, \beta, \gamma, \phi_b), \\ & \times \psi_\beta(\mathbf{O}; \beta), \psi_\gamma(\mathbf{O}; \gamma, \phi_b), \psi_{\phi_b}(\mathbf{O}; \gamma, \phi_b))^\top. \end{aligned}$$

The following proposition establishes the double robustness property and asymptotic normality of the bias corrected estimator.

Proposition 1 If either $f(Z_i, \mathbf{Z}_{N_i} | X_i, \mathbf{X}_{N_i}; \gamma, \phi_b)$ or $m(Z_i, \Sigma \mathbf{Z}_{N_i}, X_i; \beta)$ is correctly specified, then $\hat{\theta}^{\text{DR-BC}} \xrightarrow{p} \theta$ and

$$\sqrt{m}(\hat{\theta}^{\text{DR-BC}} - \theta) \xrightarrow{d} N(0, U(\theta)^{-1} V(\theta) (U(\theta)^{-1})^\top),$$

$$\text{where } U(\theta) = \mathbb{E}[-\partial \psi^{\text{DR-BC}}(\mathbf{O}_v; \theta) / \partial \theta^\top] \text{ and } V(\theta) = \mathbb{E}[\psi^{\text{DR-BC}}(\mathbf{O}_v; \theta) \psi^{\text{DR-BC}}(\mathbf{O}_v; \theta)^\top].$$

The proof of Proposition 1 can be found in [online supplementary material, Appendix F.2](#) and follows from standard estimating equation theory. The empirical sandwich variance estimator of $\hat{\theta}^{\text{DR-BC}}$ can be obtained by plugging the sample analogues of $U(\theta)$ and $V(\theta)$ in the sandwich matrix:

$$\Sigma_m = U_m(\hat{\theta})^{-1} V_m(\hat{\theta}) (U_m(\hat{\theta})^{-1})^\top,$$

where $U_m(\hat{\theta}) = m^{-1} \sum_{v=1}^m -\partial \psi^{\text{DR-BC}}(\mathbf{O}_v; \hat{\theta}) / \partial \theta^\top$ and $V_m(\hat{\theta}) = m^{-1} \sum_{v=1}^m \psi^{\text{DR-BC}}(\mathbf{O}_v; \hat{\theta}) \psi^{\text{DR-BC}}(\mathbf{O}_v; \hat{\theta})^\top$. Asymptotic normality and asymptotic variance of causal effect estimators such as $\widehat{DE}(\alpha) = \tau^\top \theta$ can be derived using the delta method, where $\tau = (-1 \ 1 \ 0 \ \dots \ 0)^\top$. For instance, the sandwich variance estimator of $\widehat{DE}(\alpha)$ is given by $\tau^\top \Sigma_m \tau$, which can be used to construct a Wald-type confidence interval for $DE(\alpha)$. The computation of the sandwich variance estimator can entail error-prone derivative calculations, owing to the presence of a component-level random effect in the treatment model (see the [online supplementary material, appendix](#) of [Perez-Heydrich et al., 2014](#)). Tedious calculations can be avoided using the R package `geex`, which computes the roots and the sandwich variance estimator for any set of unbiased estimating equations ([Saul & Hudgens, 2020](#)).

4.3 Other extensions

In [online supplementary material, Appendix A](#), we extend the regression estimator with inverse-propensity weighted coefficients (IP-WLS) considered by [Liu et al. \(2019\)](#) to the network interference setting. Instead of incorporating a correction term as in (6) and (7), inverse-propensity weighted regression is used to remove confounding and hence eliminate the bias in the regression estimators of average potential outcomes ([Kang & Schafer, 2007](#)). Contrary to what its name suggests, this estimator actually uses inverse probability-of-treatment weights, i.e. $\mathbb{1}(Z_i = z) / f(Z_i, \mathbf{Z}_{N_i} | X_i, \mathbf{X}_{N_i})$, as opposed to inverse propensity weights, i.e. $f(1, \mathbf{Z}_{N_i} | X_i, \mathbf{X}_{N_i})^{-1}$. The IP-WLS estimators of μ_{za} and μ_a are given by [online supplementary material, equations \(S1\) and \(S2\) in Appendix A](#), respectively. We show that the IP-WLS estimator has the double robustness property if errors in the outcome model are conditionally independent.

So far, the regression and DR estimators discussed have assumed independent errors, in that the outcome nuisance model does not assume a multilevel structure. Given that the Add Health data is clustered, we aimed to extend the methods to account for potential dependence in the GPAs received by students within each component. Generalizations of the above estimators to multilevel outcomes under the random intercept linear mixed model are discussed in [online supplementary material, Appendix B](#). Although these generalizations rely upon a different model to produce predicted outcome values, the form of the estimators remains unchanged. In other words, equations (4), (6), and [online supplementary material, equation \(S1\) of Appendix A](#) can still be used to respectively compute the regression estimator, the regression estimator with residual bias correction,

and the regression estimator with inverse-propensity weighted coefficients of average potential outcomes by leveraging predictions from a linear mixed model. The same is true for outcome-based estimators of marginal average potential outcomes in equations (5), (7), and [online supplementary material, equation \(S2\) of Appendix A](#).

For the most part, M-estimation of parameters is straightforward except for those involved in the IP-WLS estimator. In this case, the implementation requires weighting fixed effect coefficients by the inverse of the joint propensity score. Therefore, inference for the outcome nuisance and treatment nuisance models is performed jointly and one must be able to specify a likelihood that also accounts for the hierarchical nature of the data. More details on the implementation of weighted mixed linear models are provided in [online supplementary material, Appendix B.4](#). Contrary to the case of conditionally independent outcomes discussed above, the IP-WLS estimator does not appear to retain the double robustness property in the presence of multilevel outcomes as explained in [online supplementary material, Appendix B.4](#).

5 Simulations

We designed simulation studies to assess finite sample bias of the IPW, regression (REG), and DR estimators described in Section 4 as well as to compare their efficiency and robustness to misspecification. The first simulation scheme assumed equal component sizes and conditionally independent errors in the outcome model, while the second scheme aimed to mimic certain aspects of the Add Health study, namely the unequal network component sizes and the multilevel structure of the outcome data.

Five scenarios based on correct or incorrect specification of the outcome or treatment models, and one setting in which both models were ‘correctly specified’ but there was latent treatment dependence that was not captured by either model (see Steps 5 and 6 below). This first simulation study was conducted according to the following steps with $m = 30$ network components:

- (a) **Generation of the network:** For $v = 1, \dots, m$, we considered a fixed number of nodes $N_v = 30$. For each component, we generated the random vector H_v whose elements were a Bernoulli(0.5) distribution. The m components were generated according to an exponential random graph model (ERGM) in which matching H values increased the probability of tie between two nodes, inducing network homophily (Kolaczyk, 2009). Specifically, the ERGM used to simulate each network component was such that the conditional log-odds of two nodes having a tie while holding the rest of the network fixed was $1.5\delta_1 - 2.5\delta_2$, where δ_1 is the indicator for matching H values and δ_2 represents the increase in the number of ties. The final network G was taken as the union of the m components.
- (b) For $i = 1, \dots, N_v$, two covariates were generated as $X_{1i} \sim \mathcal{N}(0, 1)$ and $X_{2i} \sim \text{Bernoulli}(0.5)$. Denote the proportion of treated nodes in a neighbourhood as $p(z_{N_i}) = \Sigma z_{N_i} / d_i$. Potential outcomes for all $z_i \in \{0, 1\}$, $\Sigma z_{N_i} \in \{0, 1, \dots, d_i\}$ are given by

$$y_i(z_i, z_{N_i}) = 2 + 2z_i + p(z_{N_i}) + z_i p(z_{N_i}) - 1.5|X_{1i}| + 2X_{2i} - 3|X_{1i}|X_{2i} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, 1).$$

- (c) The treatment was generated with homophily as $Z_i = \text{Bernoulli}(p_i)$, where

$$p_i = \text{logit}^{-1}[0.1 + 0.2|X_{1i}| + 0.2X_{2i}|X_{1i}| - H_i + b_v], \quad b_v \sim \mathcal{N}(0, 1).$$

- (d) Based on the potential outcomes defined in Step 2, the observed outcomes were set equal to $Y_i = y_i(Z_i, \Sigma Z_j)$.
- (e) The following models were fitted to the simulated data:
 - Correct treatment model:

$$\mathbb{E}[Z_i | X_i] = \text{logit}^{-1}(\gamma_0 + \gamma_1|X_{1i}| + \gamma_2|X_{1i}|X_{2i} + \gamma_3H_i + b_v)$$

- Incorrect treatment model:

$$\mathbb{E}[Z_i | X_i] = \text{logit}^{-1}(\gamma_0 + \gamma_1 X_{1i} + \gamma_2 H_i + b_v)$$

- Correct outcome model:

$$\mathbb{E}[Y_i | X_i, Z_i, Z_{N_i}] = \beta_0 + \beta_1 Z_i + \beta_2 p(Z_{N_i}) + \beta_3 Z_i p(Z_{N_i}) + \beta_4 |X_{1i}| + \beta_5 X_{2i} + \beta_6 |X_{1i}| X_{2i}$$

- Incorrect outcome model

$$\mathbb{E}[Y_i | X_i, Z_i, Z_{N_i}] = \beta_0 + \beta_1 Z_i + \beta_2 p(Z_{N_i}) + \beta_3 Z_i p(Z_{N_i}) + \beta_4 X_{1i} + \beta_5 X_{2i}$$

- (f) Additionally, a treatment model excluding the variable driving homophily was also fitted to illustrate the threat of latent treatment homophily:

$$\mathbb{E}[Z_i | X_i] = \text{logit}^{-1}(\gamma_0 + \gamma_1 |X_{1i}| + \gamma_2 |X_{1i}| X_{2i} + b_v).$$

- (g) Estimators in (1), (4), (6) and as well as IP-WLS ([online supplementary material, equation \(S1\) of Appendix A](#)) were computed from the simulated sample for each scenario with $z \in \{0, 1\}$ and $\alpha \in \{0, 0.1, 0.2, \dots, 1\}$. The propensity score in (3) was computed using Gauss–Hermite quadrature with 10 points.

Simulations were carried out 1,000 times by repeating steps 2–7. The true parameters were calculated by averaging the potential outcomes generated in Step 2. Empirical expectations of estimators were computed by averaging the point estimates over the 1,000 replicates. The estimated standard errors were obtained from the sandwich variance matrix. Empirical coverage was defined as the proportion of simulation replicates for which the true parameters were contained in the Wald type 95% confidence intervals. Lack of convergence in the computation of the sandwich variance estimators led to the exclusion of a few samples depending on the scenario under investigation. One sample was excluded for the IPW estimator in scenarios (a) and (b), while three samples were excluded for the IPW estimator in scenarios (c) and (d). Between five and eight samples were discarded when assessing the DR-BC and IP-WLS estimators across all scenarios. No sample was excluded for the REG method.

[Figure 2](#) displays the average of the 1,000 IPW, regression (REG), the regression estimator with residual bias correction (DR-BC) and the regression estimator with inverse-propensity weighted coefficients (IP-WLS) discussed in [Section 4](#) of the direct effect $DE(\alpha)$ and indirect effect $IE(\alpha, \alpha')$, respectively, as well as the corresponding true effects. A different version of the plot in the right panel can be found in [online supplementary material, Appendix C](#) for increased readability. Unless otherwise stated, estimators are based on the correctly specified models for either treatment or outcome. As expected, the true dose response curve and the REG, DR-BC, and IP-WLS estimators almost perfectly overlap, while the IPW estimator exhibits some instability for α values closer to 1. Furthermore, the IPW estimator of $IE(\alpha, \alpha')$ departs from the true surface for larger gaps between α and α' . The IPW and REG estimators of the dose–response curve based on misspecified models show systematic bias. The DR-BC and IP-WLS estimators of $DE(\alpha)$ and $IE(\alpha, \alpha')$ based on either a misspecified treatment or a misspecified outcome model behaved as expected (results not shown). Finally, unmeasured homophily violates [Assumption 5](#) and leads to a biased IPW estimate (orange dotted curve). [Figure 3](#) shows the absolute bias of the IPW, REG, DR-BC, and IP-WLS estimators of $DE(0.6)$ under different misspecification scenarios. We see that the bias of the DR-BC and IP-WLS estimators is negligible whenever one of the two models is correctly specified. For instance, for scenario (b), the bias of IPW, REG, DR-BC, and IP-WLS are -0.011 , -0.332 , -0.014 , and -0.024 , respectively, whereas for scenario (c), these values are -0.358 , -0.004 , -0.007 , and -0.007 . In addition to exhibiting the double robustness property, the DR

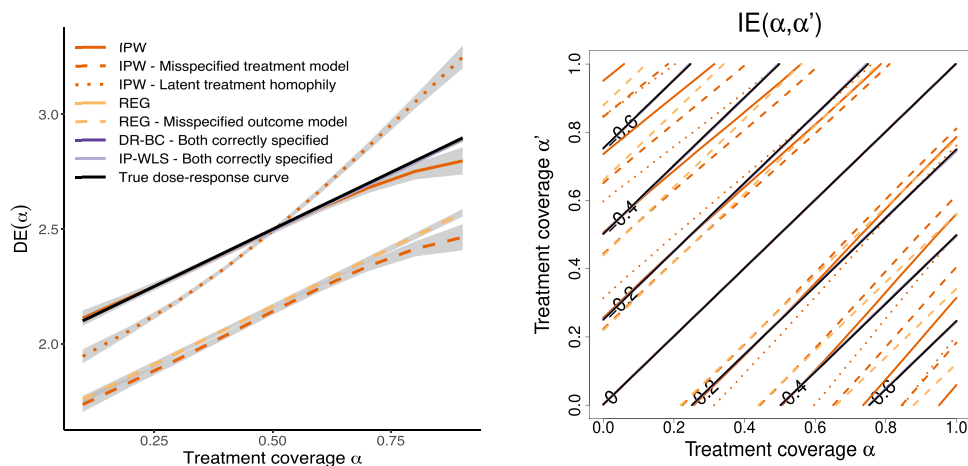


Figure 2. Left: Estimates of the dose response curve $DE(\alpha)$. Right: Estimates of the surface $IE(\alpha, \alpha')$. The black solid line represents the true effect. The light grey area represents 95% pointwise confidence intervals based on $S = 1,000$ simulation replicates.

estimator proves to be more efficient than the IPW estimator whenever at least one model of the two is correctly specified. For example, when only the model for the propensity score is correctly specified, the Monte Carlo mean squared error of the IPW estimator is 0.163 while that of the DR-BC and IP-WLS estimators are 0.032 and 0.030, respectively. However, when the outcome model is correctly specified, the REG estimator exhibits the smallest mean squared error (0.006 compared to 0.008 for both DR-BC and IP-WLS).

Empirical coverages of Wald-type 95% confidence intervals for $DE(0.6)$ based on the sandwich variance estimates can be found at the bottom of Figure 3. When either model is correctly specified, the empirical coverages of DR-BC and IP-WLS are approximately 0.95. The empirical coverages for the singly robust methods are well below 0.95 when the working model is not correctly specified. The same is true for the DR-BC and IP-WLS methods when both models are incorrect. For instance, when the treatment and outcome models are incorrect, the empirical coverages of the IPW, REG, DR-BC, and IP-WLS are 0.68, 0.43, 0.53, and 0.51, respectively. These comparisons align with what Liu et al. (2019) found in their simulation studies, albeit in a setting with partial interference only.

Online supplementary material, Table S1 in Appendix C shows the bias and the mean squared error (MSE) of the IPW, REG, DR-BC, and IP-WLS estimators of the indirect/spillover effect using the correctly specified treatment model versus the treatment model deprived of the homophilous variable, which, it should be noted, is not a confounder in the classical sense of directly impacting both treatment and outcome. In a classical causal inference setting, where units are independently and identically distributed, the omission of a predictor of the treatment that is not a confounder in the treatment nuisance model can improve the estimation (reducing the mean squared error of the estimator). However, in the network setting, these empirical results show that the IPW estimator in (1) is biased if traits shared by adjacent nodes that affect treatment assignment are unobserved or not conditioned upon.

Online supplementary material, Appendix C also contains additional simulations in which we assessed sensitivity to failure of the stratified interference assumption. In one scenario considered, the potential outcomes do not merely depend on the sum of treated neighbours but through a more complex function of the neighbourhood treatment vector and show that for this particular violation and data generating scheme, the performance of the estimators is mildly affected. In an additional scenario, we allow the potential outcomes to depend on second-order neighbours (friends of friends) and show that a misspecification of the order of the spillover effect in the outcome model leads to slight bias and undercoverage for the data generating mechanism considered.

In online supplementary material, Appendix D, we present results for a second simulation scheme in which network components were unbalanced and the outcomes were generated from

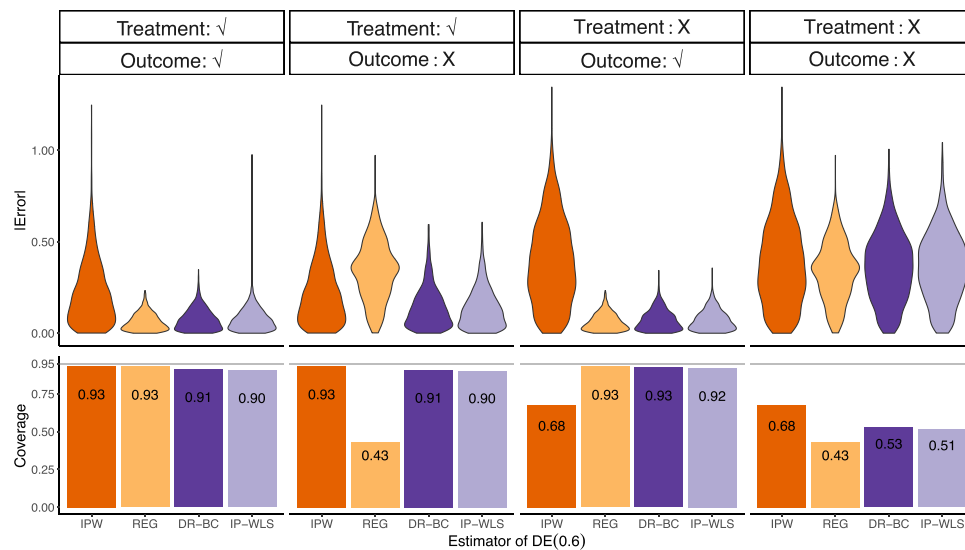


Figure 3. Absolute bias and confidence interval coverage for the IPW, REG, DR-BC, and IP-WLS estimators of $DE(0.6)$ under the first simulation scheme. In scenarios (a) and (b), results for IPW, REG, DR-BC, and IP-WLS are based on 999, 1,000, 992, and 992 simulated datasets, respectively. In scenarios (c) and (d), results for IPW, REG, DR-BC, and IP-WLS are based on 997, 1,000, 995, and 994 simulated datasets, respectively.

a random intercept linear model in order to investigate the performance of the proposed estimators under conditions similar to those encountered in the data analysis in the following section. We investigated properties of estimators for an increasingly large number of network components m , where m is from the set $\{50, 100, 150, 200\}$. This gave further empirical evidence of consistency and asymptotic normality of the singly robust methods (IPW and REG) when the correct model is used. We also gained empirical evidence of the double robustness of the DR-BC estimator for multilevel outcomes when at least one of the two models is correctly specified. Finally, when examining point estimates for the IP-WLS method, we observed a nonvanishing bias as the number of network components increased. Since there is no theoretical guarantee for the double robustness of the IP-WLS method in the presence of multilevel outcomes, we excluded this estimator from the analysis of Add Health.

6 Effect of maternal education on school performance

We re-examine a question considered by previous authors about the impact of maternal college education on adolescent school performance (Bifulco et al., 2011; Fletcher et al., 2020). In the following, the outcome is a student's GPA and the exposure is defined as the binary indicator of whether their mother has completed four years of college. The students who did not report living with their biological mother, stepmother, foster mother, or adoptive mother at the time of the survey were excluded from the analytic sample. Furthermore, we decided to exclude three schools from the analysis at outset because more than 98% of their nodes were isolates, which correspond to nodes with degree 0. This initial preparation resulted in an undirected network of 139 schools and 73,580 nodes in total.

Further pre-processing of the full sample detailed in [online supplementary material, Appendix E](#) was performed to arrive at our analytic sample. The final network structure $G = (\mathcal{N}, E)$ had 48 components (observed subnetworks) with $N = 5,724$ students and 19,776 reciprocal friendships. The size of the 48 components ranged from 2 to 508 students, with a median of 31 (first to third quartile (Q1–Q3) : 2–201). Characteristics of the network and descriptive statistics for this subsample are presented in [online supplementary material, Tables S2 and S3 of Appendix E](#), respectively. Using this analytic subsample, we fitted a linear mixed effects model for GPA and a distinct mixed effects logistic model for the propensity of maternal education at the student level. A random intercept for the network component was incorporated into each model. Given that the

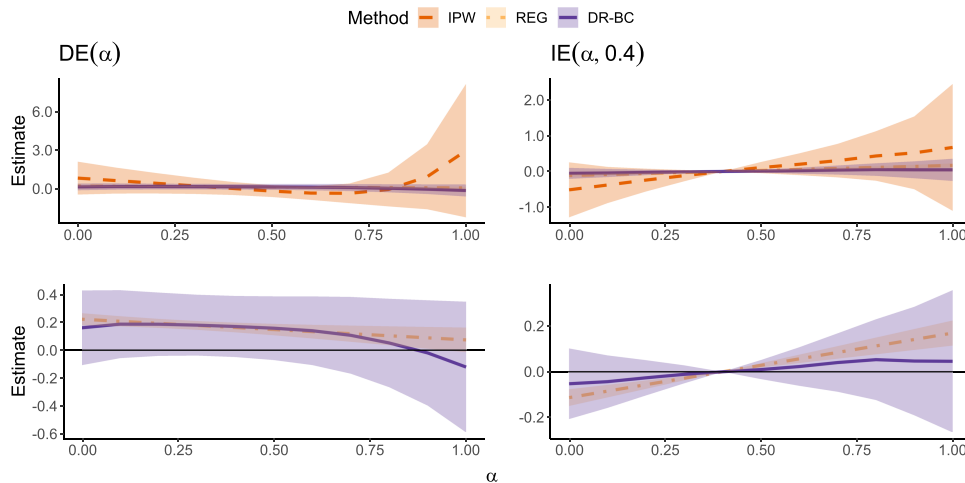


Figure 4. IPW, REG, and DR-BC estimates of $DE(\alpha)$ and $IE(\alpha, 0.4)$ effects with 95% confidence bands. The bottom panel only shows the REG and DR-BC estimates.

in-school questionnaire collected limited information about the student's mother, the model for the propensity of maternal education included only four covariates: student's race (as a proxy for the race of the mother), whether the mother was born in the U.S., whether the father is at home, and household size. Similar to Egami and Tchetgen (2023), in addition to those included in the model for the exposure, the covariates in the outcome model included gender, age, whether the student was adopted, whether the mother works for pay, perception of mother care, club participation, motivation at school, trouble at school, sense of belonging, attendance, health status, physical fitness, and screen time.

Figures 4 and 5 compare the estimators of the direct effect $DE(\alpha)$, indirect effect $IE(\alpha, 0.4)$, total effect $TE(\alpha, 0.4)$, and overall effect $OE(\alpha, 0.4)$. For all estimands, the REG and DR-BC estimators generally give similar conclusions, in that the directions of the dose-response curves align. In comparison, the dose-response curves estimated by IPW can be nonmonotonic and attain relatively high values for large α values. For instance, the REG and DR-BC estimates of $DE(\alpha)$ range between 0.22 and -0.12 and decrease with increasing coverage α , whereas the IPW estimate first decreases and increases again past $\alpha = 0.7$ to reach the value 2.95 at $\alpha = 1.0$, which is a rather extreme value considering the range of GPA (between 0 and 4). Recall that the model for maternal education was very limited compared to the outcome model, with only four covariates. Also, as shown by the histogram in online supplementary material, Figure S8 of Appendix E, the distribution of observed proportion of friends with a college-educated mother was right skewed. The reduced amount of information for higher α values is reflected in the width of the confidence bands of the IPW estimator, and in that of the DR-BC estimator to a lesser extent. Table 1 displays the IPW, REG and DR-BC estimates along with 95% confidence intervals for $DE(\alpha)$, $IE(\alpha, 0.4)$, $TE(\alpha, 0.4)$ and $OE(\alpha, 0.4)$. We provide an interpretation of $DE(0.1)$ for the DR-BC point estimate. If every student in this network were to be 'assigned' a college-educated mother versus none, we would expect the average GPA difference to be 0.18 (95% CI: $[-0.06, 0.43]$) points under a counterfactual probability of receiving treatment $\alpha = 10\%$. The interpretation of $DE(0.10)$ is nuanced, since it is not feasible for each person to be exposed and simultaneously have only 10% of their friends exposed. Recall that the target estimand $DE(\alpha)$ is defined as the expected average of individual-level direct effects standardized according to a reference population in which individuals independently receive the exposure with probability α . Both the REG and DR-BC methods yield a decreasing $DE(\alpha)$ curve, although the point estimates given by DR-BC never reach statistical significance. This could mean that there is little to no added benefit for a student to have a college-educated mother, and even less so if there is already a high proportion of 'exposed' friends in their social circle.

Using the DR-BC method, we obtain an estimate of -0.04 (95% CI: $[-0.16, 0.07]$) for $IE(0.1, 0.4)$. This implies that if we held the maternal education indicator to 0, the GPA of students

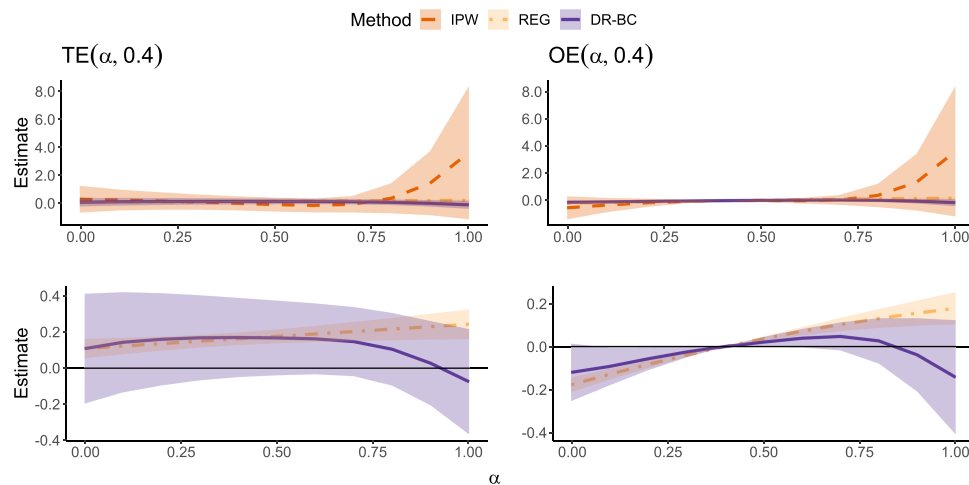


Figure 5. IPW, REG, and DR-BC estimates of $TE(\alpha, 0.4)$ and $OE(\alpha, 0.4)$ with 95% confidence bands. The bottom panel only shows the REG and DR-BC estimates.

with 10% of friends with college-educated mothers would be on average lower by -0.04 points compared to those with 40% of friends exposed, assuming both groups' covariate distribution matched that of the overall population. The REG method suggests a small positive spillover effect of maternal education on GPA, as $IE(\alpha, 0.4)$ increases with α . This aligns with some findings of Fletcher et al. (2020). However, the DR-BC method does not suggest such an effect, as the estimates do not significantly differ from 0. Since the total effect is the sum of the direct and indirect effects, it remains relatively constant across different α values. Similarly to the indirect effect, the point estimate of the overall effect, which marginalizes over the individual exposure/treatment, increases with the proportion of treated friends for values up to $\alpha = 0.7$. For instance, the DR-BC estimates are -0.09 (95% CI: $[-0.18, -0.01]$), -0.03 (95% CI: $[-0.05, 0.01]$), and 0.05 (95% CI: $[-0.02, 0.11]$) for $OE(0.1, 0.4)$, $OE(0.3, 0.4)$, and $OE(0.7, 0.4)$, respectively.

7 Discussion

In this article, we propose DR estimators for causal effects in the presence of clustered network interference. They are shown to be consistent and asymptotically normal if either the treatment model or the outcome regression model is correctly specified. In simulations, we generated networks with equal and unequal component sizes and outcome data with and without a multilevel structure. For conditionally independent outcomes, empirical results illustrate the DR property and the potential efficiency gain of the DR-BC and IP-WLS estimators over previously proposed IPW estimators when the propensity score model is correctly specified. Results also underscore the importance of the no latent treatment homophily assumption for the identification of causal effects, an assumption that we have made explicit in our framework. If the outcome is assumed to follow a hierarchical model, then the DR-BC estimator remains DR, while evidence suggests that IP-WLS is only consistent if the outcome model is correctly specified. It is worth noting that the proposed methods can be applied to observational network data that do not have a clear multilevel structure from the outset. Indeed, community detection algorithms can be used to identify clusters beforehand, as was done under partial interference (Liu et al., 2019; Perez-Heydrich et al., 2014) and networked interference (Lee et al., 2023). Using causal structural equation models, Ogburn et al. (2022) proposed a TMLE approach for a single social network. However, they use a bounded dependence central limit theorem to establish asymptotic properties, which precludes the presence of nodes with high degree centrality or so-called 'hubs' in the social network. That is, conditions for asymptotic normality will not hold if the degree of one or more nodes grows faster than $\sqrt{N}$ (Ogburn et al., 2022). Our methods are free of such assumptions, as our

Table 1. IPW, REG, and DR-BC estimates for the add health study along with 95% Wald-type confidence intervals for $DE(\alpha)$, $IE(\alpha, 0.4)$, $TE(\alpha, 0.4)$ and $OE(\alpha, 0.4)$

Estimand	IPW	REG	DR-BC
$DE(\alpha)$			
$\alpha = 0.1$	0.64 [−0.36,1.64]	0.20 [0.17,0.24]	0.18 [−0.06,0.43]
$\alpha = 0.3$	0.24 [−0.39,0.86]	0.18 [0.14,0.21]	0.18 [−0.04,0.39]
$\alpha = 0.5$	−0.17 [−0.62,0.29]	0.15 [0.11,0.19]	0.16 [−0.07,0.38]
$\alpha = 0.7$	−0.34 [−1.10,0.42]	0.12 [0.06,0.17]	0.11 [−0.17,0.38]
$IE(\alpha, 0.4)$			
$\alpha = 0.1$	−0.38 [−0.88,0.13]	−0.08 [−0.11,−0.06]	−0.04 [−0.16,0.07]
$\alpha = 0.3$	−0.12 [−0.27,0.04]	−0.03 [−0.04,−0.02]	−0.01 [−0.05,0.03]
$\alpha = 0.5$	0.11 [−0.05,0.26]	0.03 [0.02,0.04]	0.01 [−0.03,0.05]
$\alpha = 0.7$	0.30 [−0.16,0.76]	0.08 [0.06,0.11]	0.04 [−0.09,0.17]
$TE(\alpha, 0.4)$			
$\alpha = 0.1$	0.26 [−0.48,1.01]	0.12 [0.08,0.17]	0.14 [−0.13,0.42]
$\alpha = 0.3$	0.12 [−0.42,0.66]	0.15 [0.11,0.18]	0.17 [−0.07,0.4]
$\alpha = 0.5$	−0.06 [−0.55,0.43]	0.17 [0.14,0.21]	0.17 [−0.04,0.37]
$\alpha = 0.7$	−0.04 [−0.62,0.55]	0.20 [0.15,0.25]	0.15 [−0.04,0.33]
$OE(\alpha, 0.4)$			
$\alpha = 0.1$	−0.32 [−0.85,0.20]	−0.13 [−0.15,−0.10]	−0.09 [−0.18,−0.01]
$\alpha = 0.3$	−0.06 [−0.20,0.09]	−0.04 [−0.05,−0.03]	−0.03 [−0.05,0.01]
$\alpha = 0.5$	0.01 [−0.10,0.13]	0.04 [0.03,0.05]	0.02 [−0.01,0.04]
$\alpha = 0.7$	0.05 [−0.30,0.40]	0.10 [0.07,0.13]	0.05 [−0.02,0.11]

asymptotic framework views each network component as a datum and holds regardless of the network topology within each component.

Our study of peer characteristics in education leveraging the Add Health dataset did not provide evidence of a spillover effect of maternal education on academic performance within adolescents' social circles. However, when outcome regression was used, we did find that a higher proportion of friends with a college-educated mother led to slightly higher GPA among students whose mothers had not completed a 4-year college degree, corroborating some substantive conclusions of Fletcher et al. (2020). This study is nonetheless subject to important limitations. First of all, the exposure of interest, maternal education, was not well defined and did not correspond to a standard intervention, potentially violating Assumption 1. Also, the exposure and outcome were contemporaneous, making it essentially impossible to distinguish effects due to interference and homophily or environmental confounding (Egami & Tchetgen, 2023; VanderWeele & An, 2013). In their causal peer effect analysis of Add Health, Egami and Tchetgen (2023) proposed a double negative control approach to adjust for potential network confounding and introduced a time element by using GPA outcomes reported in the in-school and the in-home interviews, which were separated by a period of at least 90 days. However, it is known that the Add Health network at baseline changed overtime. Methods such as stochastic actor-oriented models could be better adapted to this longitudinal setting, as they model social influence and selection of ties jointly (Snijders, 2005; VanderWeele & An, 2013). Two other limitations worth mentioning were the need to select a subsample of the network for computational reasons, and the use of single imputation. The first of these was largely practical but did lead to some meaningful differences in the covariate distribution of the subsample compared with that of the full sample, affecting generalizability of the results.

While our analysis assumes first-order spillover effects only, this assumption could be relaxed by widening the interference set to include second-order neighbours (friends of friends). In some settings it could be desirable to allow the treatment assignments received by friends of friends to have

a smaller impact than those received by immediate neighbours. One avenue for future work would be to allow spillover effects to decay with the neighbourhood order, which would require a more substantial extension. Nonetheless, we assessed finite sample properties of the estimators under a violation of first-order spillover effects via simulation and observed that bias and coverage were not severely affected even when the data were generated under a second-order interference mechanism.

Methodological work on causal inference in the presence of network interference is still in its infancy, and there are several areas of future research related to the counterfactual framework presented in this article. One natural route would be to extend this framework to longitudinal network data, which would allow to address network endogeneity by disentangling effects from social influence and homophily (Tchetgen & VanderWeele, 2012). Future work could entail formulating stochastic actor-oriented models within a counterfactual framework. Furthermore, in this article, the observed social network is assumed to be undirected and considered as known and fixed, such that inference is entirely driven by random treatment assignment and random potential outcomes (Li & Wager, 2022). However, it is possible that subjects misreport their connections and that social ties are missing not at random (VanderWeele & An, 2013). While the general problem of imputation of network data has received some attention (see Huisman, 2009; Krause et al., 2018), to our knowledge, the only existing imputation procedure for causal inference under network interference was proposed in the context of experiments (Kao, 2017). This could be another important avenue for future work in the area of observational network-based studies. Existing causal inference methods could also possibly be extended to the classes of directed networks and weighted networks, wherein the direction (respectively strength) of edges would be of importance.

A partially observed network, that is, a network with uncertain or missing edges, would imply that the interference sets are not known with certainty which would have an impact on the estimation of spillover effects. In the experimental setting, Li et al. (2021) formulated the impact of network error on the bias of estimators of average causal effects. The impact of such errors in observational network-based studies has yet to be studied. In addition, asymptotic results are derived under the assumption that the known interference graph can be partitioned into several components (Lee et al., 2023; Liu et al., 2016). Forastiere et al. (2021) assumed that the social network is a random sample from a larger underlying network and devised bootstrapping procedures based on egocentric and cluster sampling. Future research could involve relaxing the fixed network assumption and assume that the interference graph is a random draw from an (unknown) graphon as was done in experimental settings (Li & Wager, 2022). Such an inference framework would point to a clearer use of resampling methods for network data to derive finite-sample variance estimators for causal effects.

In the no interference setting, many types of DR estimators have been proposed. Notably, the class of DR estimators with propensity-based covariates remains to be investigated in the network interference setting. For instance, Liu et al. (2019) formulated a generalization of the estimator proposed by Scharfstein et al. (1999) to the partial interference setting, but other estimators such as the one proposed by Bang and Robins (2005, 2008) could also be considered. Lastly, while Park and Kang (2022) showed that the bias-corrected DR estimator of Liu et al. (2019) was locally efficient under partial interference (Zhang et al., 2023), whether the proposed estimators are semi-parametric efficient has yet to be investigated in the network interference setting.

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Data availability

The data from Add Health used in this study were provided by the Carolina Population Center and are not publicly available. Researchers can request access at <https://data.cpc.unc.edu/projects/2/view>. The simulation code and datasets are available from the corresponding author on reasonable request. Codes and a sample dataset can be found on github:

<https://github.com/vanessamcnealis/Network-Interference-estimators>.

Supplementary material

Supplementary material is available online at *Journal of the Royal Statistical Society: Series C*.

Appendix

List of Figures

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